



Online Marketplace for Short-Term Accommodations

Project Proposal – Academic Year 2025

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1. Supervisor/Advisor Details

	Supervisor 1 (IT Related)	Supervisor 2 (Optional)
Name		
Designation		
Workplace address		
Academic/Professional Qualifications		
Work Experience		
Email		
Phone Number		

2. Project Details

3. Group Members' Details

Name	Index Number	Registration Number	Role
Jayaweera K.U.E	FC213008	FC110620	Project Manager
Nishanka N.M.R.S	FC213018	FC110630	System Analyst
Dinushanth S.	FC213034	FC110661	System Designer
Wickramasinghe W.M.D.D	FC213029	FC110641	Database Administrator
Gunasekara M.P.I.S	FC213003	FC110615	Frontend developer
De Silva W.P.Y.G	FC213039	FC110986	Backend developer
Zeenathsharoth M.Y	FC213013	FC110625	Backend developer
Nizmy M.N.M	FC213023	FC110635	Tester / QA

4. Introduction

ShortStay is a web-based platform designed to connect travelers with local hosts offering short-term accommodations.

The platform simplifies the booking experience and enables Sri Lankan residents to earn additional income by renting out rooms or properties.

The system will feature listings, bookings, payments, reviews, and an admin panel for platform moderation.

5. Motivation for the Project

Sri Lanka, a top tourist destination, is recovering from economic challenges.

Many locals have unused space that could generate extra income through short-term rentals.

However, international platforms are not always user-friendly for local users.

Our project, ShortStay, aims to bridge this gap by providing a localized, accessible solution that benefits both tourists and Sri Lankan communities.

6. Objectives of the Project

- Create a user-friendly web application for short-term accommodation bookings
- Allow hosts to list and manage properties
- Enable guests to search and book places with secure payment
- Collect feedback and ratings from users
- Provide an admin interface for system oversight and compliance

7. Scope of the Project

- Focuses on guest-host interactions for short-term stays
- No integration with government licensing or taxation initially
- System will be responsive and optimized for browsers
- Future modules (messaging, maps, etc.) can be added in phases

8. User Classes and Characteristics

- Guest – Traveler looking for temporary stays
- Host – Property owner offering accommodation
- Admin – System operator for user and listing verification
- Payment Manager – Responsible for managing all payment-related activities and financial reporting
- Field Inspector – Visits host properties, checks conditions, and verifies standard compliance for badge approval

9. Product Perspective

ShortStay will be built as a responsive web application. It integrates with payment gateways and email APIs for communication.

10. Design and Implementation Constraints

- Must support all modern web browsers
- Requires active internet connection
- Data must be securely stored and transferred (HTTPS)

11. Assumptions and Dependencies

- Users are familiar with basic web navigation
- APIs used for payment and email are assumed reliable

12. Specific Requirements

Core Features:

- User login with role-based access (Guest, Host, Admin, Payment Manager, Field Inspector)
- Listing creation and management (Hosts)
- Search and filter properties (Guests)
- Booking system with date selection and confirmation
- Notifications via Email
- Review and rating mechanism
- Payment processing and financial management (Payment Manager)
- Property inspection and verification system (Field Inspector)

Guest Functionalities:

- Search and book accommodations
- Manage bookings and view history
- Leave feedback and rate hosts

Host Functionalities:

- List and edit properties with images and availability
- View and manage bookings
- Respond to guest reviews

Admin Functionalities:

- Verify host and guest registrations
- Approve/reject property listings
- View system logs and handle complaints

Payment Manager Functionalities:

- Monitor and manage all incoming/outgoing transactions
- Generate monthly financial reports
- Handle disputes related to payments

Field Inspector Functionalities:

- Visit host locations and inspect facilities
- Evaluate host properties based on set criteria
- Approve or reject verification badge requests

13. Critical Functionalities

- Role-based Authentication
- Advanced search and filters
- Calendar-integrated booking system

- Real-time email notifications
- Admin compliance tools
- Review and feedback system

14. Deliverables

- Fully functional web system
- Database with data models and relationships
- Technical and user documentation
- Project presentation and demo
- User support materials

15. Resource Requirements

Software Requirements:

- HTML, CSS, JavaScript (React)
- Node.js or PHP backend
- MySQL or MongoDB
- Visual Studio Code / Atom
- Email API (e.g., SendGrid)
- Google Chrome / Firefox

Hardware Requirements:

- Intel Core i3 or higher
- Minimum 4GB RAM (8GB recommended)
- 256GB SSD or higher
- Stable internet connection

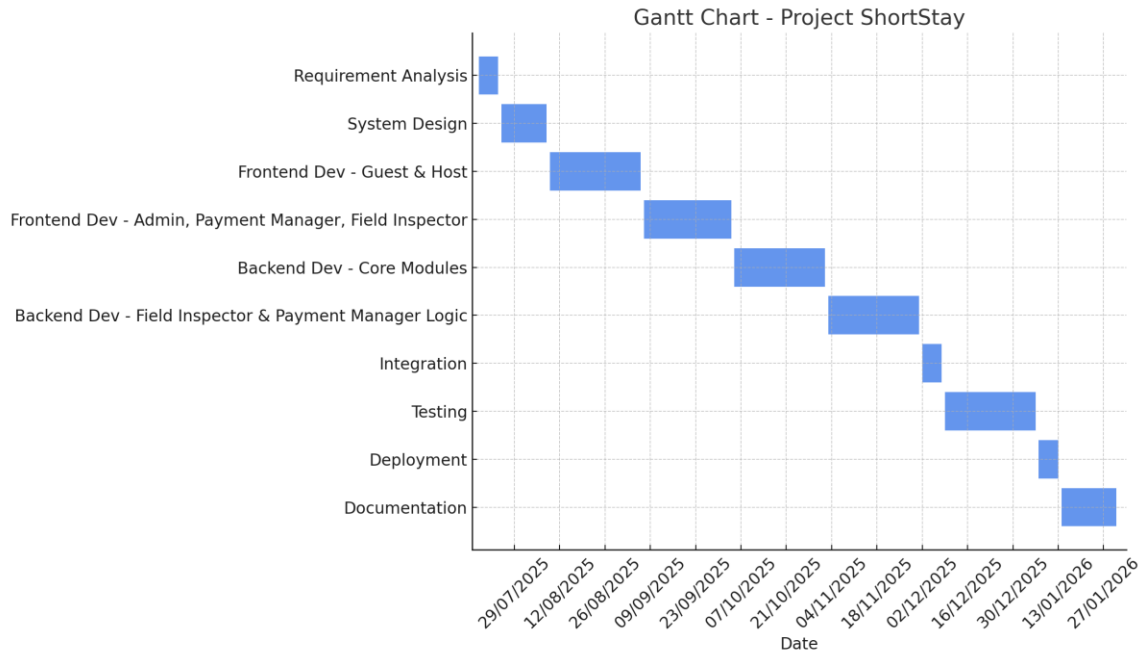
16. Proposed Method for Evaluating Success

- Unit testing for each module
- Integration testing across modules
- System-wide functional testing
- Feedback from sample users
- Supervisor review and approval

17. Proposed Client/Beneficiary

This project is developed for academic purposes at the University of Sri Jayewardenepura. However, it has potential to benefit Sri Lankan communities by promoting local tourism and creating new income opportunities through short-term accommodation sharing.

18. Gantt Chart



-THE END-